10) Write a Java program to demonstrate safe thread suspension, resumption, and termination using a control flag and interrupt() instead of the deprecated suspend(), resume(), and stop() methods.

PROGRAM:

```
class MyThread extends Thread {
  private volatile boolean running = true;
  private volatile boolean suspended = false;

  public void run() {
    try {
      while (running) {

        synchronized (this) {
          while (suspended) {
            wait();
          }
        }

        System.out.println("Thread is running...");
        Thread.sleep(500);
      }
    } catch (InterruptedException e) {
      System.out.println("Thread interrupted");
    }

    System.out.println("Thread terminated.");
  }
```

```java
    public synchronized void suspendThread() {
        suspended = true;
        System.out.println("Thread suspended.");
    }

    public synchronized void resumeThread() {
        suspended = false;
        notify();
        System.out.println("Thread resumed.");
    }

    public void stopThread() {
        running = false;
        interrupt();
    }
}

public class Main {
    public static void main(String[] args) throws InterruptedException {

        MyThread t = new MyThread();

        t.start();

        Thread.sleep(2000);

        t.suspendThread();
```

```java
        Thread.sleep(2000);

        t.resumeThread();

        Thread.sleep(2000);

        t.stopThread();

        t.join();

        System.out.println("Main thread completed.");
    }
}
```

OUTPUT:

Thread is running...

Thread is running...

Thread is running...

Thread is running...

Thread suspended.

Thread resumed.

Thread is running...

Thread is running...

Thread is running...

Thread is running...

Thread interrupted

Thread terminated.

Main thread completed.